



Scuola Superiore Sant'Anna – Ericsson
Novel Modulation Formats for Optical Short Reach Interconnect

High Speed Optical Transmission for 5G Radio Access

Technical Report

Enrico Forestieri
Marco Secondini
Marco Romagnoli
Vito Sorianello
Gabriele De Angelis
Michele Midrio
Alessandro Nottola
Marco Chiesa
Luca Potì
Francesco Fresi
Gianluca Meloni

November 2017

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A. Unipolar M -ary PAM Signaling

A.1. Thermal-Noise Regime

OOK is one of the simplest modulation formats and is widely deployed in optical fiber communications due to the simplicity of the required modulation and detection schemes. It can be obtained by directly modulating a laser (at low bit rates) or using an external amplitude modulator and it can be easily detected by using a simple direct-detection scheme. However, even when considering such a simple detection technique, OOK signaling is not the only possible option. Different (possibly multilevel) modulation formats can also be considered, provided that the different waveforms associated to different modulation symbols can be reliably detected after photodetection (in particular, after the square-modulus operation). For instance, when considering a generic linearly-modulated optical signal with complex envelope

$$x(t) = \sum_k a_k g(t - kT) \quad (\text{A.1})$$

where T is the symbol time, a_k is the k -th transmitted symbol belonging to the M -ary alphabet $\mathcal{A} = \{A_1, A_2, \dots, A_M\}$, and $g(t)$ the pulse shape (which we assume to vanish outside symbol-time interval T to avoid nonlinear ISI after photodetection), reliable detection is possible only if all the modulation symbols have different energy (i.e., without loss of generality, $|A_1|^2 < |A_2|^2 < \dots < |A_M|^2$). Therefore, it is not possible to directly detect PSK or conventional (bipolar) pulse amplitude modulation (PAM) formats, as the phase or sign information of each modulation symbol is lost after photodetection. However, it is possible to correctly detect a unipolar M -ary PAM—a simple multilevel extension of the OOK modulation format. When thermal noise dominates, i.e., when the noise is independent of the signal, in order to minimize the BER, the modulation symbols are to be chosen such that to give rise to equispaced symbols after photodetection, and thus $A_m = A\sqrt{m-1}$, $m = 1, \dots, M$.

Neglecting the shot-noise and denoting by $h_e(t)$ is the impulse response of the electrical filter (modeling the response of the photodetector as well as any post-detection filter), the received photocurrent can be written as

$$i(t) = [i'(t) + n(t)] \otimes h_e(t) \quad (\text{A.2})$$

where $n(t)$ is real AWGN modeling the effect of electronics noise, and

$$i'(t) = \mathcal{R}|x(t)|^2 = \mathcal{R} \sum_k |a_k|^2 |g(t - kT)|^2 \quad (\text{A.3})$$

is the noiseless electrical signal *before* filtering, $\mathcal{R}$ being the responsivity of the photodetector. The power spectral density of $n(t)$ is usually expressed in terms of the equivalent noise current density of the photodetector i_n [A/ $\sqrt{\text{Hz}}$] as $N_0/2 = i_n^2/2$. Note that $i'(t)$ is a real PAM signal, with symbols $c_k = |a_k|^2$ and pulse shape $p(t) = |g(t)|^2$, affected by AWGN. Therefore, optimum detection can be implemented by matched filtering, symbol-time sampling, and threshold detection, with thresholds properly chosen to minimize the error probability.

In order to compute the system performance, it is useful to define the energy of the optical pulse

$$\mathcal{E}_g \triangleq \int_{-\infty}^{\infty} |g(t)|^2 dt \quad (\text{A.4})$$

and the energy of the electrical pulse

$$\mathcal{E}_p \triangleq \int_{-\infty}^{\infty} |p(t)|^2 dt = \int_{-\infty}^{\infty} |g(t)|^4 dt \quad (\text{A.5})$$

From (A.2) and (A.3), assuming $h_e(t) = |g(-t)|^2$, we can express the received samples as $i_k = i(kT) = s_k + n_k$, where

$$s_k = \mathcal{R}|a_k|^2 \mathcal{E}_p \quad (\text{A.6})$$

is the signal component, while the noise component n_k is zero-mean Gaussian with variance

$$\sigma_{\text{elec}}^2 = i_n^2 B_e = \frac{i_n^2}{2} \mathcal{E}_p \quad (\text{A.7})$$

independent of the signal component, B_e being the electronics noise equivalent bandwidth. Finally, dropping index k for simplicity of notation, decisions are taken by comparing the received sample $i = s + n$ with thresholds γ_m according to

$$\hat{a} = \begin{cases} A_1, & i < \gamma_1 \\ A_m, & \gamma_{m-1} < i < \gamma_m, \quad m = 2, \dots, M-1 \\ A_M, & i > \gamma_{M-1} \end{cases} \quad (\text{A.8})$$

Assuming equiprobable symbols $A_m = A\sqrt{m-1}$, we note that the distance between adjacent levels is

constant and equal to

$$d \triangleq \mathcal{R}A^2\mathcal{E}_p \quad (\text{A.9})$$

Moreover, given the symmetry of the conditional distributions (they are Gaussian with same variance), optimum detection thresholds are located in the middle between adjacent signal levels, i.e., at

$$\gamma_m = \frac{\mathcal{R}A^2}{2}(2m-1)\mathcal{E}_p, \quad m = 1, \dots, M-1 \quad (\text{A.10})$$

The corresponding symbol error probability (SER) is

$$\begin{aligned} \text{SER} &= \frac{1}{M} \left[P(i > \gamma_1 | a = A_1) + P(i < \gamma_{M-1} | a = A_M) + \sum_{m=2}^{M-1} P(i < \gamma_{m-1} \cup i > \gamma_m | a = A_m) \right] \\ &= \frac{1}{M} \left[P(n > d/2) + P(n < d/2) + \sum_{m=2}^{M-1} P(n > d/2) + P(n < d/2) \right] \\ &= \frac{2(M-1)}{M} Q\left(\frac{d}{2\sigma_{\text{el}}}\right) = \frac{2(M-1)}{M} Q\left(\sqrt{\frac{\mathcal{R}^2 A^4 \mathcal{E}_p}{2i_n^2}}\right) \end{aligned} \quad (\text{A.11})$$

where

$$Q(x) \triangleq \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-y^2/2} dy = \frac{1}{2} \text{erfc}\left(\frac{x}{\sqrt{2}}\right) \quad (\text{A.12})$$

Finally, assuming a Gray mapping of bits on symbols, the bit error probability can be approximated as (with high probability, only one bit is wrong per each wrong symbol)

$$\text{BER} \simeq \frac{2(M-1)}{M \log_2 M} Q\left(\sqrt{\frac{\mathcal{R}^2 A^4 \mathcal{E}_p}{2i_n^2}}\right) \quad (\text{A.13})$$

Now, in order to obtain a fair comparison among different modulation formats, we consider a fixed bit rate $R_b = 1/T_b$ and, therefore, a symbol time T that scales proportionally to the number of encoded bits per symbol

$$T = T_b \log_2 M \quad (\text{A.14})$$

Moreover, we compute

$$E\{|a_k|^2\} = \frac{A^2}{M} \sum_{m=1}^{M-1} m = \frac{A^2(M-1)}{2} \quad (\text{A.15})$$

$$E\{|a_k|^4\} = \frac{A^4}{M} \sum_{m=1}^{M-1} m^2 = \frac{A^4(M-1)(2M-1)}{6} \quad (\text{A.16})$$

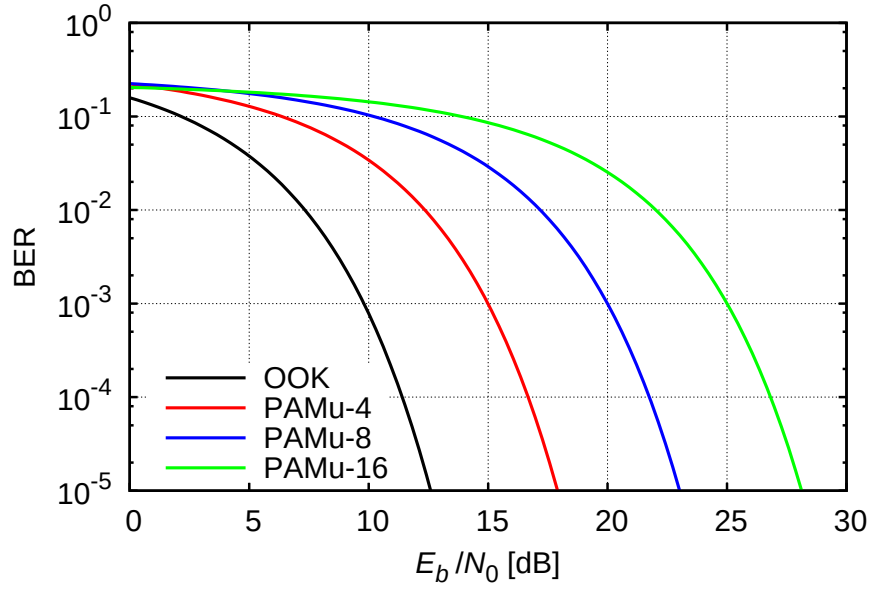


Fig. A.1.: Bit-error probability of M -ary unipolar PAM as a function of the *electrical* SNR per bit.

and eventually express the quantities appearing in (A.13) in terms of optical or electrical power (or energy per bit). Given the restrictions on the duration of the pulse shape $g(t)$, the average *optical* power can be expressed as

$$P_o = E\{|a_k|^2\} \frac{\mathcal{E}_g}{T} = \frac{A^2 \mathcal{E}_g (M-1)}{2T} \quad (\text{A.17})$$

while the average *electrical* power (of the unfiltered signal in (A.3)) can be expressed as

$$P_e = \mathcal{R}^2 E\{|a_k|^4\} \frac{\mathcal{E}_p}{T} = \frac{\mathcal{R}^2 A^4 \mathcal{E}_p (M-1)(2M-1)}{6T} \quad (\text{A.18})$$

Using (A.14) and (A.18), the error probability can be related to the electrical power as

$$\text{BER} \simeq \frac{2(M-1)}{M \log_2 M} Q \left(\sqrt{\frac{3 \log_2 M}{(M-1)(2M-1)} \cdot \frac{P_e T_b}{i_n^2}} \right) \quad (\text{A.19})$$

As expected from standard results in digital communications, (A.19) depends only on the electrical SNR per bit (the last factor under the square root, i.e., $P_e T_b / i_n^2 = E_b / N_0$) and on the number of modulation symbols M , while it is independent of the pulse shape. Fig. A.1 reports the error probability as a function of the electrical SNR per bit for different values of the alphabet size M . The additional electrical power (electrical power penalty) required to increase the alphabet size from $M = 2$ (OOK) to $M = 4$ (PAM-4) at an error probability of 10^{-3} is about 5.2 dB while, for large values of M , the

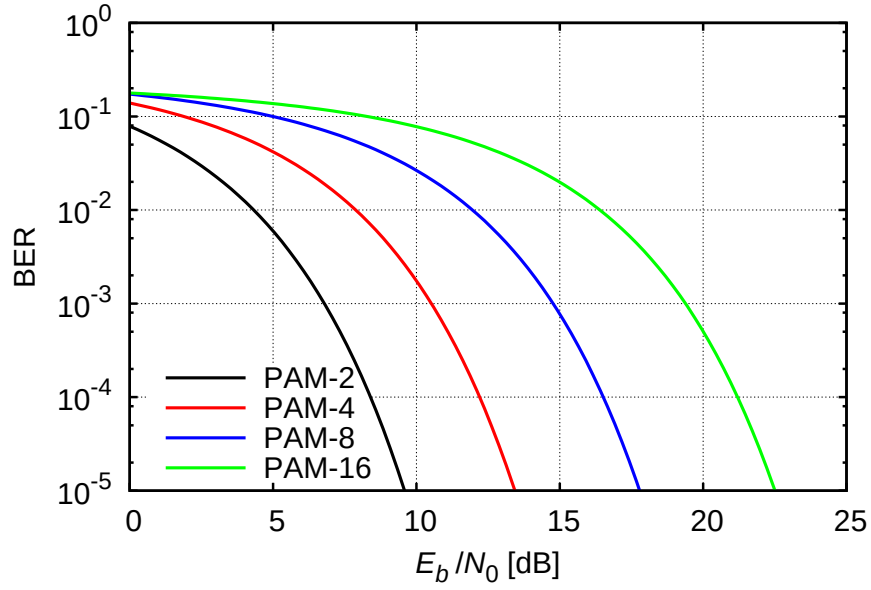


Fig. A.2.: Bit-error probability of M -ary PAM as a function of the electrical SNR per bit (useful only for comparison, as this format is not compatible with a direct-detection scheme).

electrical power penalty for doubling the alphabet size (adding one more bit per symbol) converges to 6 dB.

It is interesting to compare (A.19) to the error probability obtained for a conventional bipolar PAM [35]

$$\text{BER} \simeq \frac{2(M-1)}{M \log_2 M} Q \left(\sqrt{\frac{6 \log_2 M}{(M^2-1)} \cdot \frac{P_e T_b}{i_n^2}} \right) \quad (\text{A.20})$$

that is reported in Fig. A.2. It can be easily verified that the advantage of bipolar PAM over unipolar PAM is 3 dB for $M = 2$ (antipodal signaling versus OOK) and increases up to 6 dB for large M . Moreover, for bipolar PAM, the electrical power penalty for doubling the alphabet size is lower than for unipolar PAM at low values of M (while it converges to the same value of 6 dB for large M). In fact, the penalty of bipolar PAM-4 compared to bipolar PAM-2 at an error probability of 10^{-3} is about 3.73 dB

On the other hand, if we want to refer the performance to the optical power, it is useful to measure the efficiency with which the optical power is converted into electrical power. Taking into account (A.4) and (A.5), from (A.17) and (A.18) we have the following ratio

$$\frac{P_e}{P_o^2} = \mathcal{R}^2 \cdot \frac{E\{|a_k|^4\}}{[E\{|a_k|^2\}]^2} \cdot \frac{T \int_{-\infty}^{\infty} |g(t)|^4 dt}{[\int_{-\infty}^{\infty} |g(t)|^2 dt]^2} = \mathcal{R}^2 \cdot f_A \cdot f_g \quad (\text{A.21})$$

For a constant-envelope signal, both efficiency factors f_A and f_g are equal to 1 and the overall conversion efficiency depends only on the photodetector responsivity $\mathcal{R}$. For our unipolar PAM modulation alphabet, from (A.15) and (A.16) we have

$$f_A \triangleq \frac{E\{|a_k|^4\}}{[E\{|a_k|^2\}]^2} = \frac{2(2M-1)}{3(M-1)} \quad (\text{A.22})$$

that is a *modulation* efficiency factor $f_A = 2$ for OOK, slowly decreasing to $f_A = 4/3$ for $M \rightarrow \infty$. Moreover, for ideal rectangular pulses $g(t) = \text{rect}(t/\tau)$ of duration $\tau \leq T$, we have

$$f_g \triangleq \frac{T \int_{-\infty}^{\infty} |g(t)|^4 dt}{[\int_{-\infty}^{\infty} |g(t)|^2 dt]^2} = \frac{T}{\tau} \quad (\text{A.23})$$

that is a *pulse-shape* efficiency factor $f_g = 1$ for NRZ pulses, linearly increasing with the inverse of the pulse width for RZ pulses. Using (A.21), we can eventually express the bit-error probability (A.19) as

$$\begin{aligned} \text{BER} &\simeq \frac{2(M-1)}{M \log_2 M} Q \left(\sqrt{\frac{3 \log_2 M}{(M-1)(2M-1)} \cdot \frac{\mathcal{R}^2 f_A f_g P_o^2 T_b}{i_n^2}} \right) \\ &= \frac{2(M-1)}{M \log_2 M} Q \left(\sqrt{\frac{\log_2 M}{(M-1)^2} \cdot \frac{2 \mathcal{R}^2 f_g P_o^2 T_b}{i_n^2}} \right) \end{aligned} \quad (\text{A.24})$$

We note several important differences between the equivalent expressions (A.19) and (A.24). First of all, the argument of the square root inside the Q-function scales *linearly* with the electrical power but *quadratically* with the optical power. Moreover, the dependence on the number of symbols M is slightly different, due to the modulation efficiency factor f_A . The combination of these two factors is responsible for a significantly different scaling of receiver sensitivity (in terms of required *optical* or *electrical* power) with the numbers of modulation symbols M . Finally, in the optical case also the pulse shape plays an important role through the efficiency factor f_g (such that a better performance is obtained by using narrower RZ pulses), while it is completely irrelevant in the electrical case. Fig. A.3 reports the bit error probability as a function of the normalized optical power

$$\frac{\mathcal{R} P_o \sqrt{2 f_g T_b}}{i_n} = \frac{E_{b,o} \sqrt{2 f_g}}{\text{NEP} \sqrt{T_b}} \quad (\text{A.25})$$

for different alphabet sizes M . In (A.25), $E_{b,o} = P_o T_b$ is the average optical energy per bit, while $\text{NEP} = i_n / \mathcal{R}$ is the so called *noise equivalent power* in $[\text{W}/\sqrt{\text{Hz}}]$, usually defined as the optical

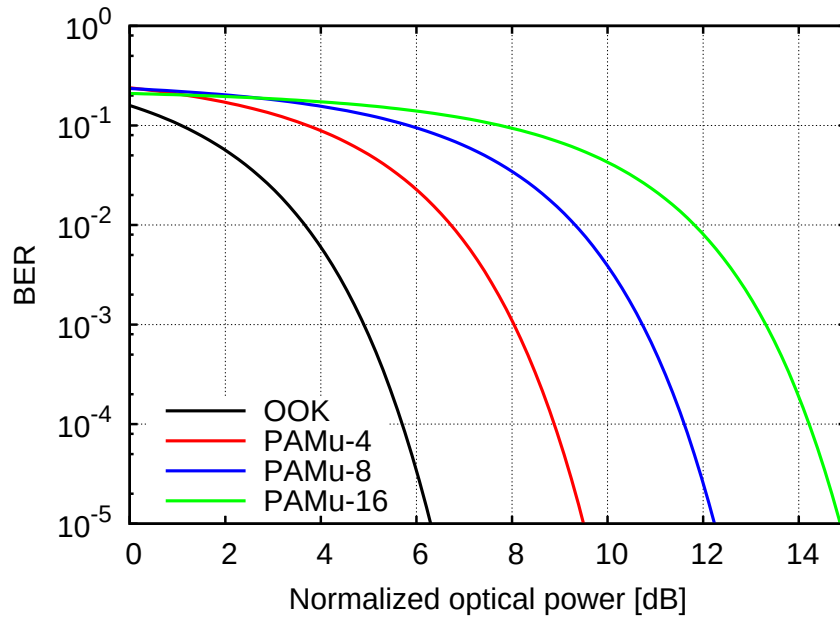


Fig. A.3.: Bit-error probability of M -ary unipolar PAM as a function of the normalized *optical* power.

power per square-root electrical receiver bandwidth that would be required to make the electrical power equal to the electronics noise variance. While the computation of the *absolute* performance (required optical power) requires, in this case, to specify some additional physical parameters, it is nevertheless possible from (A.24) or Fig. A.3 to compute the *relative* performance for different values of M , i.e., the optical power penalty due to an increase of the alphabet size M . The additional optical power required to increase the alphabet size from $M = 2$ (OOK) to $M = 4$ (PAM-4) at an error probability of 10^{-3} is about 3.14 dB. The optical power penalty for doubling the alphabet size (adding one more bit per symbol) remains about 3 dB also for different values of M and converges asymptotically to this exact value for large M .

It is interesting to compare Fig. A.1 with Fig. A.3, providing the same error probability as a function of two different quantities—the first one proportional to the electrical power; the second one proportional to the optical power. It clearly turns out that increasing the alphabet size is less expensive in terms of optical power than in terms of electrical power.

A.2. ASE-Noise Regime

Optical preamplification can be used to boost a weak received signal to appreciable optical power levels prior to detection. At the same time, ASE noise with a power spectral density per polarization mode of $N_{\text{ASE}} = (G - 1)n_{\text{sp}}h\nu$ is introduced by the amplification process. Since the ASE spectrum

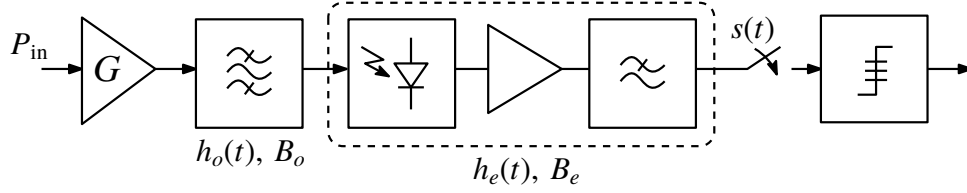


Fig. A.4.: Block diagram of a preamplified receiver.

is much broader than the signal spectrum, an optical bandpass filter is employed to suppress out-of-band ASE, as shown in Fig. A.4. Upon detection, the random ASE field beats against the signal field (as well as against itself), leading to signal-ASE beat noise and ASE-ASE beat noise after electrical filtering. The photocurrent generated at the detector can be written as

$$s(t) = \left[\Re \{ \sqrt{G} x(t) + n_{\text{ASE}}(t) \}^2 + \Re \{ n'_{\text{ASE}}(t) \}^2 + i_{\text{shot}}(t) + i_{\text{elec}}(t) \right] \otimes h_e(t) \quad (\text{A.26})$$

where $x(t)$ is the complex envelope (A.1) of the signal after the optical filter, while $n_{\text{ASE}}(t)$ and $n'_{\text{ASE}}(t)$ are the complex envelopes of the ASE noise in the same polarization as the signal and the orthogonal polarization, respectively, again after optical filtering. In (A.26), $i_{\text{shot}}(t)$ and $i_{\text{elec}}(t)$ are the shot-noise and thermal electronics noise currents, respectively. The average value and variance of the photocurrent $s(t)$ are

$$\begin{aligned} E\{s(t)\} &= \Re(GP_s + 2P_{\text{ASE}}) \\ \sigma^2 &= \sigma_{\text{elec}}^2 + \sigma_{\text{shot}}^2 + \sigma_{\text{s-ASE}}^2 + \sigma_{\text{ASE-ASE}}^2 \end{aligned} \quad (\text{A.27})$$

where $P_s = E\{|x(t)|^2\} \approx P_{\text{in}}$, $P_{\text{ASE}} = E\{|n_{\text{ASE}}(t)|^2\} = E\{|n'_{\text{ASE}}(t)|^2\}$, σ_{elec}^2 is the electronics noise variance (A.7) and, letting

$$r_n(\tau) = E\{n_{\text{ASE}}^*(t)n_{\text{ASE}}(t+\tau)\}, \quad r_e(\tau) = \int_{-\infty}^{\infty} h_e(t)h_e(t-\tau)dt,$$

the remaining three terms are

$$\begin{aligned} \sigma_{\text{shot}}^2 &= 2q[\Re(GP_s + P_{\text{ASE}})]B_e \\ \sigma_{\text{s-ASE}}^2(t) &= 2\Re^2 G \Re \left\{ \iint_{-\infty}^{\infty} x(\tau_1)x^*(\tau_2)r_n(\tau_1-\tau_2)h_e(t-\tau_1)h_e(t-\tau_2)d\tau_1d\tau_2 \right\} \\ \sigma_{\text{ASE-ASE}}^2 &= 2\Re^2 \int_{-\infty}^{\infty} |r_n(\tau)|^2 r_e(\tau)d\tau \end{aligned} \quad (\text{A.28})$$

where we assumed $h_e(t)$ normalized such that $\int_{-\infty}^{\infty} h_e(t)dt = 1$, and denoted by q the electron charge.

In the limit of rectangular filters and constant input power P_{in} to the optical preamplifier, the previous relations simplify to¹

$$\begin{aligned}\sigma_{\text{shot}}^2 &= 2q[\mathcal{R}(GP_s + P_{\text{ASE}})]B_e \\ \sigma_{\text{s-ASE}}^2 &\approx 4\mathcal{R}^2 GP_s N_{\text{ASE}} B_e \\ \sigma_{\text{ASE-ASE}}^2 &\approx 4\mathcal{R}^2 N_{\text{ASE}}^2 B_o B_e\end{aligned}\tag{A.29}$$

Neglecting P_{ASE} with respect to GP_s and replacing $N_{\text{ASE}} = (G - 1)n_{\text{sp}}h\nu$ and $\mathcal{R} = \eta q/h\nu$, where $\eta \leq 1$ is the quantum efficiency, we obtain

$$\begin{aligned}\sigma_{\text{shot}}^2 &\approx 2q^2 \eta GP_s B_e / h\nu \\ \sigma_{\text{s-ASE}}^2 &\approx 4(q\eta G)^2 n_{\text{sp}} P_s B_e / h\nu \\ \sigma_{\text{ASE-ASE}}^2 &\approx 4(q\eta G n_{\text{sp}})^2 B_o B_e\end{aligned}\tag{A.30}$$

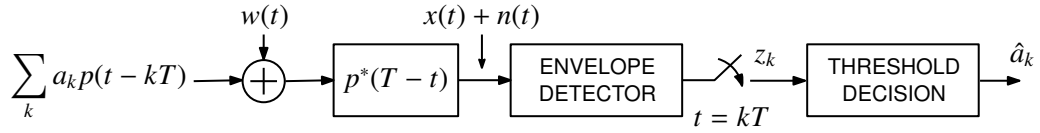
As can be seen, the variance of the optical beat noise is proportional to the square of the optical gain G^2 , so that the optical noise dominates shot noise as

$$\frac{\sigma_{\text{s-ASE}}^2}{\sigma_{\text{shot}}^2} \approx 2\eta G n_{\text{sp}} \gg 1\tag{A.31}$$

Once the gain of the optical amplifier is chosen high enough to let optical beat noise also dominate electronics noise, the main impact on receiver performance comes from optical and electrical filter characteristics. In the absence of ISI, optimum receiver performance is achieved if the optical filter is matched to the data pulses, and if the electrical filter is made sufficiently broadband to have no influence on the detected signal. However, the electrical bandwidth is usually upper-bounded by technological constraints and cost considerations, especially at very high data rates, and ISI often does have a significant impact on detection. It is nevertheless interesting to know what is the optimal performance.

Neglecting shot-noise, electronics noise, and—for the time being—the ASE noise in the orthogonal polarization, it is easy to see that the optimal performance corresponds to that of asynchronous

¹In this case, the power spectral density of $n_{\text{ASE}}(t)$ is equal to N_{ASE} for $-B_o/2 \leq f \leq B_o/2$, so that $r_n(\tau) = N_{\text{ASE}} B_o \text{sinc}(B_o \tau)$, while $r_e(\tau) = h_e(\tau) \otimes h_e(-\tau) = h_e(\tau) = 2B_e \text{sinc}(2B_e \tau)$, because $h_e(\tau)$ is ideal lowpass with bandwidth B_e . For constant input power, $x(\tau_1)x^*(\tau_2) \approx P_s$, while, taking into account that $B_o > B_e$ and thus $\int_{-\infty}^{\infty} r_n(\tau_1 - \tau_2) h_e(t - \tau_1) d\tau_1 = \int_{-\infty}^{\infty} r_n(\tau) h_e(t - \tau_2 - \tau) d\tau = N_{\text{ASE}} h_e(t - \tau_2)$, we have $\iint_{-\infty}^{\infty} r_n(\tau_1 - \tau_2) h_e(t - \tau_1) h_e(t - \tau_2) d\tau_1 d\tau_2 = N_{\text{ASE}} \int_{-\infty}^{\infty} h_e^2(t - \tau_2) d\tau_2 = 2N_{\text{ASE}} B_e$, which accounts for $\sigma_{\text{s-ASE}}^2$. It can also be shown that, for $B_o > B_e$, $\int_{-\infty}^{\infty} |r_n(\tau)|^2 r_e(\tau) d\tau = 2N_{\text{ASE}}^2 B_o^2 B_e \int_{-\infty}^{\infty} \text{sinc}^2(B_o \tau) \text{sinc}(2B_e \tau) d\tau = 2N_{\text{ASE}}^2 B_o B_e (1 - B_e/2B_o) \approx 2N_{\text{ASE}}^2 B_o B_e$, which accounts for $\sigma_{\text{ASE-ASE}}^2$.


 Fig. A.5.: Block diagram of asynchronous detection of unipolar PAM- M .

(incoherent) detection of unipolar PAM- M in additive white Gaussian noise, as illustrated in Fig. A.5, as we are going to explain. Here, in order to have an exact correspondence to our optical system, *i*) the filter impulse response $p^*(T - t)$ matched to the PAM pulse $p(t)$ corresponds to the optical filter impulse response $h_o(t)$ such that $p(t) \otimes h_o(t) = g(t)$ and thus the complex envelope $x(t)$ of the signal component after the optical filter is as in (A.1) and (A.26); *ii*) $w(t)$ is AWGN with power spectral density $N_0 = N_{\text{ASE}}/G = (G - 1)n_{\text{sp}}h\nu/G \approx h\nu$, such that the filtered noise is $n(t) = n_{\text{ASE}}(t)/\sqrt{G}$, where $n_{\text{ASE}}(t)$ is as in (A.26). We note that, apart the inessential different scaling, when in Fig. A.4 $h_o(t) = p^*(T - t)$ and $h_e(t)$ is sufficiently broadband, the samples at the detector correspond to the square of the samples in Fig. A.5, so that when the threshold levels in Fig. A.4 correspond to the square of the threshold levels in Fig. A.5, the performance of the two systems is the same. The advantage of using the block diagram in Fig. A.5 lies in the fact that it allows for a simple analytical approximation for the bit error rate, as we will see. Conditional upon $a_k \in \{A_1, A_2, \dots, A_M\}$, the sample z_k in Fig. A.5 has the Ricean distribution [35]

$$f_{z_k}(z|a_k = A_i) = \frac{2z}{\mathcal{E}_p N_0} \exp\left(-\frac{z^2 + A_i^2 \mathcal{E}_p^2}{\mathcal{E}_p N_0}\right) I_0\left(\frac{2A_i z}{N_0}\right), \quad z \geq 0 \quad (\text{A.32})$$

where

$$\mathcal{E}_p \triangleq \int_{-\infty}^{\infty} |p(t)|^2 dt \quad (\text{A.33})$$

Note that if $A_i = 0$, which is a typical choice for A_1 , (A.32) becomes a Rayleigh distribution [35]. In (A.32), $I_0(x)$ is the modified Bessel function of the first kind and order zero, which for $x \gg 1$ can be approximated as $I_0(x) \approx e^x/\sqrt{2\pi x}$. Hence, for $A_i/N_0 \gg 1$, (A.32) simplifies to

$$f_{z_k}(z|a_k = A_i) \approx \frac{\sqrt{z/(A_i \mathcal{E}_p)}}{\sqrt{\pi \mathcal{E}_p N_0}} \exp\left(-\frac{(z - A_i \mathcal{E}_p)^2}{\mathcal{E}_p N_0}\right), \quad z \geq 0 \quad (\text{A.34})$$

which is an almost Gaussian shape with mean $A_i \mathcal{E}_p$ and variance $\mathcal{E}_p N_0/2$.

The decision on the transmitted symbol is taken by comparing the sample z_k with $M - 1$ thresholds

γ_m according to the rule

$$\hat{a}_k = \begin{cases} A_1, & z_k < \gamma_1 \\ A_m, & \gamma_{m-1} < z_k < \gamma_m, \quad m = 2, \dots, M-1 \\ A_M, & z_k > \gamma_{M-1} \end{cases} \quad (\text{A.35})$$

Correspondingly, assuming equiprobable symbols, the SER can be evaluated as

$$\text{SER} = \frac{1}{M} \left[P(z_k > \gamma_1 | a_k = A_1) + P(z_k < \gamma_{M-1} | a_k = A_M) + \sum_{m=2}^{M-1} P(\{z_k < \gamma_{m-1}\} \cup \{z_k > \gamma_m\} | a_k = A_m) \right] \quad (\text{A.36})$$

where each term is the error probability conditional to a given symbol. These error probabilities are given by the area under the tail(s) of the pdf corresponding to the given threshold(s). Only one tail contributes to the error probability for $a_k = A_1$ and $a_k = A_M$, while both left and right tails contribute in all other cases. The optimum thresholds are determined by the crossings of two adjacent pdfs, as this guarantees that (A.36) is minimized.

We are now ready to analyze the impact of the chosen alphabet on the SER. As already seen, when thermal noise dominates, in order to minimize the SER, the modulation symbols are to be chosen such that to give rise to equispaced levels after photodetection, and thus $A_m = \sqrt{m-1}A'$, $m = 1, \dots, M$. This is because, in this case, the conditional pdfs at the detector are Gaussian with same variance, so that the optimum thresholds are midway between adjacent signal levels. However, when ASE noise dominates this is not true anymore. In this case, from (A.34) we see that it is the electrical field amplitude that is almost Gaussian distributed with a variance independent of the transmitted symbol (possibly with the exception of the symbol A_1). Hence, for minimizing the SER, the symbols are to be chosen as $A_m = (m-1)A$, $m = 1, \dots, M$. This can also be explained graphically as follows.

The average power of the received signal in Fig. A.5 corresponds to the received optical power at the input of the optical preamplifier and can be expressed as

$$P_{\text{in}} = E\{a_k^2\} \frac{\mathcal{E}_p}{T} = \begin{cases} \frac{(M-1)A'^2 \mathcal{E}_p}{2T} & \text{if } a_k \in \{\sqrt{m-1}A'\}_{m=1}^M \\ \frac{(M-1)(2M-1)A^2 \mathcal{E}_p}{6T} & \text{if } a_k \in \{(m-1)A\}_{m=1}^M \end{cases} \quad (\text{A.37})$$

Choosing $A' = \sqrt{(2M-1)/3}A$ in order to have the same average power, the conditional pdfs corre-

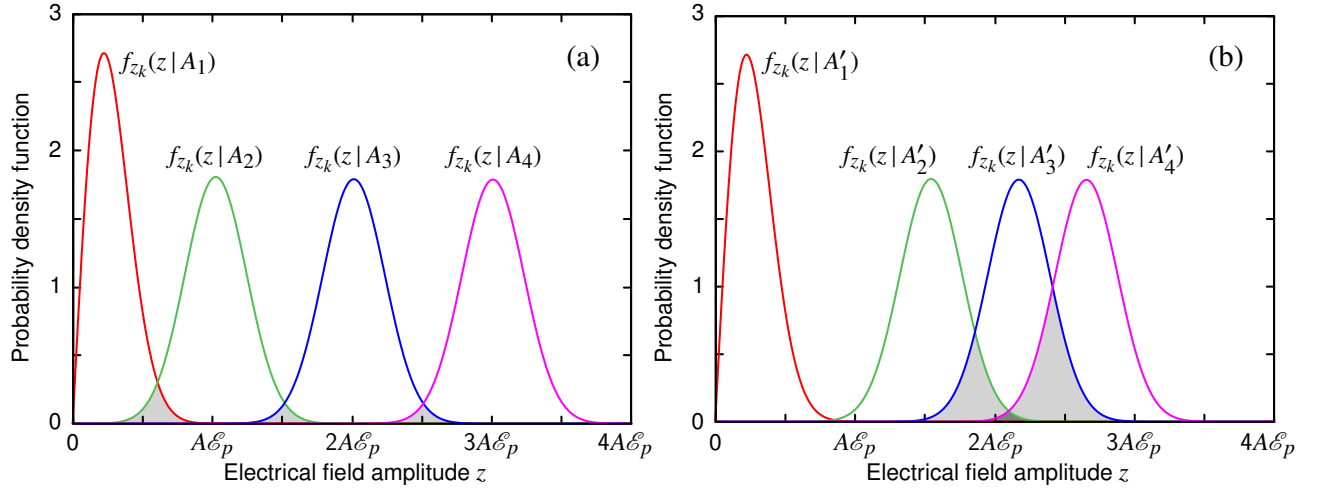


Fig. A.6.: Conditional pdfs for (a) equispaced amplitudes ($A_m = (m-1)A$, $m = 1, \dots, M$) and (b) equispaced intensities ($A'_m = \sqrt{(m-1)(2M-1)/3}A$, $m = 1, \dots, M$) for $M = 4$.

sponding to the same noise level are shown in Fig. A.6 for both cases and $M = 4$. As can be seen, except for the error probability conditional on $a_k = A'_1$, all other conditional error probabilities are larger with equispaced intensities. In this case, it also turns out that the average error probability, which according to (A.36) is equal to the shaded area divided by M , is larger (note that the smaller and darker shaded area in Fig. A.6(b) has to be counted twice).

In the limit of the validity of (A.34), with good approximation the optimum thresholds γ_m in (A.35) are midway between adjacent signal levels, with the only possible exception of γ_1 , which might be slightly larger. However, we are going to disregard this small difference and develop an expression for the BER that will be later shown to be very accurate when also taking into account the ASE noise in the orthogonal polarization and the exact optimum thresholds. Thus, using

$$\begin{aligned} A_m &= (m-1)A \\ \gamma_m &= (m-1/2)A\mathcal{E}_p \end{aligned} \quad m = 1, \dots, M$$

the conditional error probabilities in (A.36) turn out to be

$$P(z_k > \gamma_1 | a_k = A_1) = \int_{A\mathcal{E}_p/2}^{\infty} \frac{2z}{\mathcal{E}_p N_0} \exp\left(-\frac{z^2}{\mathcal{E}_p N_0}\right) dz = \exp\left(-\frac{A^2 \mathcal{E}_p}{4N_0}\right) \quad (\text{A.38})$$

$$\begin{aligned}
P(z_k < \gamma_{M-1} | a_k = A_M) &= 1 - \int_{(M-\frac{3}{2})A\sqrt{\frac{2\mathcal{E}_p}{N_0}}}^{\infty} \frac{2z}{\mathcal{E}_p N_0} \exp\left(-\frac{z^2 + (M-1)^2 A^2 \mathcal{E}_p^2}{\mathcal{E}_p N_0}\right) I_0\left(\frac{2(M-1)Az}{N_0}\right) dz \\
&= 1 - \int_{(M-\frac{3}{2})A\sqrt{\frac{2\mathcal{E}_p}{N_0}}}^{\infty} x \exp\left(-\frac{x^2 + 2(M-1)^2 A^2 \mathcal{E}_p / N_0}{2}\right) I_0\left((M-1)A\sqrt{\frac{2\mathcal{E}_p}{N_0}}x\right) dx \\
&= 1 - Q_1\left((M-1)A\sqrt{\frac{2\mathcal{E}_p}{N_0}}, (M-\frac{3}{2})A\sqrt{\frac{2\mathcal{E}_p}{N_0}}\right) \tag{A.39}
\end{aligned}$$

$$\begin{aligned}
P(\{z_k < \gamma_{m-1}\} \cup \{z_k > \gamma_m\} | a_k = A_m) &= 1 - Q_1\left((m-1)A\sqrt{\frac{2\mathcal{E}_p}{N_0}}, (m-\frac{3}{2})A\sqrt{\frac{2\mathcal{E}_p}{N_0}}\right) + Q_1\left((m-1)A\sqrt{\frac{2\mathcal{E}_p}{N_0}}, (m-\frac{1}{2})A\sqrt{\frac{2\mathcal{E}_p}{N_0}}\right) \tag{A.40}
\end{aligned}$$

where $Q_1(a, b)$ is the first-order Marcum Q -function [35], [36], defined as

$$Q_1(a, b) \triangleq \int_b^{\infty} x \exp\left(-\frac{x^2 + a^2}{2}\right) I_0(ax) dx. \tag{A.41}$$

For $b \gg 1$ and $b \gg b - a$ the first-order Marcum Q -function can be approximated by the standard Q -function

$$Q(x) \triangleq \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-y^2/2} dy = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right) \tag{A.42}$$

through the asymptotic expansion $Q_1(a, b) \simeq Q(b - a)$ [36]. It is easy to verify that the above conditions hold in our case, so that we can use the following approximations for $m \geq 2$

$$1 - Q_1\left((m-1)A\sqrt{\frac{2\mathcal{E}_p}{N_0}}, (m-\frac{3}{2})A\sqrt{\frac{2\mathcal{E}_p}{N_0}}\right) \simeq Q\left(\sqrt{\frac{A^2 \mathcal{E}_p}{2N_0}}\right) \tag{A.43}$$

$$Q_1\left((m-1)A\sqrt{\frac{2\mathcal{E}_p}{N_0}}, (m-\frac{1}{2})A\sqrt{\frac{2\mathcal{E}_p}{N_0}}\right) \simeq Q\left(\sqrt{\frac{A^2 \mathcal{E}_p}{2N_0}}\right) \tag{A.44}$$

where (A.43) stems from the fact that in this case $b - a < 0$ and $Q(x) = 1 - Q(-x)$. Taking into

account (A.43) and (A.44), and replacing (A.38)–(A.40) into (A.36), we obtain

$$\text{SER} \simeq \frac{1}{M} \left[\exp\left(-\frac{A^2 \mathcal{E}_p}{4N_0}\right) + (2M-3)Q\left(\sqrt{\frac{A^2 \mathcal{E}_p}{2N_0}}\right) \right] \quad (\text{A.45})$$

From (A.37), the average energy per symbol is $E_s = \log_2 M \cdot E_b = P_{\text{in}} T = (M-1)(2M-1)A^2 \mathcal{E}_p / 6$, E_b being the average energy per bit. Hence, replacing

$$A^2 \mathcal{E}_p = \frac{6 \log_2 M}{(M-1)(2M-1)} E_b \quad (\text{A.46})$$

in (A.45) and assuming Gray coding, so that $\text{BER} \approx \text{SER} / \log_2 M$, we have

$$\text{BER} \simeq \frac{1}{M \log_2 M} \left[\exp\left(-\frac{3 \log_2 M}{2(M-1)(2M-1)} \frac{E_b}{N_0}\right) + (2M-3)Q\left(\sqrt{\frac{3 \log_2 M}{(M-1)(2M-1)} \frac{E_b}{N_0}}\right) \right] \quad (\text{A.47})$$

Taking into account (A.42), (A.47) corresponds to (1.10). As $N_0 \approx h\nu$, the ratio E_b/N_0 can be interpreted as the number of photons per bit at the input of the optical preamplifier.

We recall that (A.47) holds under the hypotheses of no ISI, matched optical filter, and no post-detection electrical filter. In particular, in the presence of a narrowband electrical filter, no exact closed form expression is known for the BER. However, in this case, the BER can be computed by the saddle-point approximation method, which requires the knowledge of the moment generating function (MGF) of the photodetected sample [37]. This MGF can be obtained by a proper Karhunen-Loève expansion of the received signal, and the ISI arising from transmission impairments can also be accounted for (see [30] for a survey of the available methods and details). The saddle-point method is very versatile and can also be applied to our idealized case. It suffices considering $p(t)$ in Fig. A.5 as an ideal NRZ pulse of width T , so that the matched filter is simply an integrate&dump filter. The BER obtained through (A.47) for $M = 4$ is reported in Fig. A.7 as the curve labeled “analytical expression”. We found that (A.47) is an excellent approximation, as the curve obtained by using the exact expression with the Marcum Q -function is indistinguishable from it. Also the results obtained through the saddle-point approximation turned out to be indistinguishable from (A.47) when using the same thresholds. However, recall that (A.47) was obtained by also neglecting the ASE noise in the polarization orthogonal to the signal. As accounting for it is easy when using the saddle-point method, we did so and obtained the blue curve in Fig. A.7, showing that this noise component introduces a penalty of about 0.5 dB. Then, instead of using fixed thresholds placed midway between adjacent signal levels, we computed the BER by optimizing them. Surprisingly enough, we obtained a curve

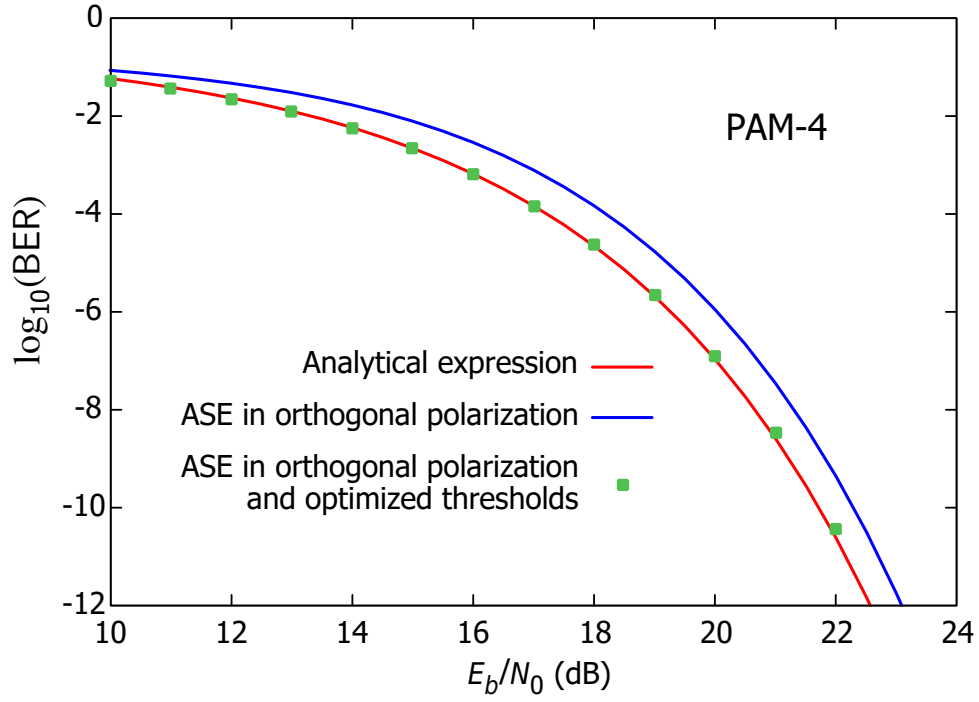


Fig. A.7.: Performance of PAM-4 in the absence of ISI and matched filtering.

that is barely distinguishable from (A.47), so that we report it as a series of square dots in Fig. A.7. In conclusion, (A.47) turns out to be an excellent approximation when taking into account the ASE noise in the orthogonal polarization and the thresholds are optimized.

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